

Greedy algorithms

Greedy algorithms are heuristics which make decisions which are the best at the moment.

1. Write a program solving the travelling salesman problem in a very simple way – always choose the vertex which is the closest one to the current position. Can you propose an instance where this strategy leads to a suboptimal solution?
2. Try to create a greedy algorithm for the knapsack problem. Prove that this approach does not return exact solutions.
3. You have a set of activities, each starts at a certain time and ends at a certain time (for example you can play tennis from 12:00 to 13:00). Your task is to choose as many of them as possible, but they cannot overlap. Can you find a greedy algorithm which is a correct solution to this problem?